

by the standard deviation of that variable. A noteworthy characteristic of a standardized variable is that its mean is consistently zero, and its standard deviation always equals 1.

5.9 LIMITATIONS OF REGRESSION ANALYSIS

Even when we possess theoretically sound variables that can explain a phenomenon and we manage to gather data on these variables, it is a common challenge not to have a clear understanding of the functional relationship between the dependent variable and the independent variables (see data generation process in Unit 3). In two-variable models, the scatter diagram frequently allows for an assessment of the functional relationship. However, in multiple regression models, identifying the suitable functional form becomes challenging, as visually interpreting scatter diagrams in multiple dimensions is not feasible. Consequently, the stochastic disturbances u_i play a crucial role in regression analysis, a significance we will delve into as we advance to subsequent units. Undoubtedly, selecting a suitable model for empirical estimation demands a considerable amount of skill and experience. However, certain guidelines can be provided:

The literature and theoretical paradigms must be re-visited as the foundational theory might propose a specific type of functional form. It is advisable to determine both the rate of change (slope) of the dependent variable concerning the independent variable and the elasticity of the dependent variable concerning the independent variable.

The parameters selected for the model should align with specific *a priori* expectations. For instance, when examining the demand for laptops in relation to price and other variables, it is reasonable to anticipate a negative coefficient for the price variable.

Occasionally, a given set of data may be reasonably well-fitted by more than one model. R squared values may help in deciding which model is better in some such cases. However, there are some caveats to be considered – as we will explore in the upcoming sections, the inclusion of more regressors in the model leads to an increase in R^2 . What holds greater significance is the theoretical foundation of the chosen model, the signs of the estimated coefficients, and its statistical significance? If the performance of a model is better on the basis of these criteria, a lower value of R^2 does not matter much.

When considering the limitations of regression analysis, several important factors should be taken into account:

Linearity Assumption: It is crucial to assess whether the relationship between the variables under investigation truly follows a linear pattern. This assumption must be carefully examined because deviations from linearity can lead to inaccurate conclusions.

Sensitivity to Outliers: Regression analysis can be highly sensitive to outliers, which are data points that deviate significantly from the overall trend. Researchers must exercise caution when interpreting results, as outliers can

disproportionately influence regression coefficients and may not accurately represent the underlying relationship.

Preliminary Data Visualization: Prior to conducting regression analysis, it is advisable to visualize the data through plots and graphs. This step helps identify potential outliers and provides insights into the overall data distribution. Decisions regarding outlier handling and data transformations should be made based on this preliminary exploration.

Extrapolation Risks: Extrapolation or making predictions outside the range of the explanatory variable, should be approached with caution. Extending regression results beyond the observed data range can lead to unreliable predictions and is often not appropriate. For example, using regression to predict outcomes for a population significantly different from the study sample may yield inaccurate results.

Hidden Variables and Confounding Factors: Researchers should be aware of the presence of lurking variables, also known as confounding or hidden variables. These unaccounted-for factors can distort the relationship between the variables included in the regression model. When confounding variables are present, the observed relationship may be misleading or spurious. Careful consideration and control of potential confounders are essential in regression analysis.

In summary, regression analysis offers valuable insights into relationships between variables, but it is not without limitations. Researchers must critically assess the linearity assumption, address the impact of outliers, visualize data before analysis, exercise caution with extrapolation, and remain vigilant for hidden variables to ensure the reliability and validity of their regression results.

Check Your Progress 2

- 1) Explain how you can visually detect the presence of heteroscedasticity in regression analysis. What are the characteristics of a residual plot indicating heteroscedasticity.

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- 2) Write down the visual methods for detecting autocorrelation in time series or regression models. How do autocorrelation function plots and lag plots help in identifying autocorrelation?

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- 3) Under what circumstances would you consider omitting the intercept term in a regression model?

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- 4) Discuss the implications of a zero-intercept model and the differences in interpretation compared to models with intercepts.

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5.10 LET US SUM UP

Ensuring accurate model specification is crucial in econometric analysis, as it allows for a focused exploration of relevant factors impacting the research theme. Challenges inevitably arise in the quest for a perfect econometric model, underscoring the need for accurate parameter estimation and thorough validation checks, to uphold the underlying assumptions.

Residual analysis emerges as a pivotal tool in assessing model adequacy, with various patterns in residual plots offering valuable insights into model fit. By scrutinizing these patterns, issues such as heteroscedasticity and autocorrelation can be identified and subsequently addressed in an effective manner. Furthermore, the impact of outliers and variable units on regression coefficients is highlighted in the unit. Conducting tests to detect outliers, assess normality, and handle specific scenarios in regression analysis, such as regression through the origin, adjusting variable origins and scales, and selecting the most suitable functional form, significantly contributes to accurate model diagnostics. Finally, certain potential challenges in regression analysis are discussed. By navigating these challenges and leveraging diagnostic techniques, economists can enhance the robustness and reliability of their regression analyses.

5.11 KEY WORDS

- Residual** : It refers to the difference between the observed value and the predicted value in regression analysis.
- Homoscedasticity** : It refers to the presence of uniform variance of residuals across all values of the independent variable.

Model Diagnostics	: Techniques for assessing the adequacy and assumptions of regression models.
Normality	: It refers to the assumption that the error term in a regression model follows a normal distribution.
Endogeneity Problem	: It occurs when an explanatory variable in a regression model is correlated with the error term, violating the assumption of exogeneity.
Functional Form	: The mathematical expression that defines the relationship between the dependent and independent variables in an economic model.
Economic Model	: It refers to the representation of relationship between a response variable (dependent variable) and the predictor variable(s) (independent variable(s)), typically specified by subject or area experts to capture relevant economic phenomena.

5.12 SOME USEFUL BOOKS/REFERENCE

Gujarati, D. (2014). *Econometrics by Examples*. Bloomsbury Publishing.

Verbeek, M. (2017). *A guide to modern econometrics*. John Wiley & Sons.

Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*. MIT Press.

Links:

Berman H.B., "Residual Analysis in Regression". Available at: <https://stattrek.com/regression/residual-analysis> [Accessed on Date: 9/6/2023].

Econometric Data Analysis: A Cost-Effective, Largely Successful Methodology By Lee S. Friedman. Available at: <https://gsppi.berkeley.edu/lee-friedman/documents/econometric-data-analysis-2.pdf>

IIT, Kanpur: <http://home.iitk.ac.in/~shalab/regression/Chapter4-Regression-ModelAdequacyChecking.pdf>

5.13 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) You should mention the impact of influential observations on estimation results and provide examples of situations where such observations might occur. Also, touch upon the concept of robust estimation techniques and their relevance in mitigating these issues (Refer to Section 5.1 & Section 5.2)

**Classical Regression
Model**

- 2) Refer to the patterns in residual plots and how they indicate the presence of specific problems.
- 3) Think about the influence outliers can have on the estimation process and the potential biases they introduce into regression results.

Check Your Progress 2

- 1) Consider the patterns in residual plots and how they differ between homoscedastic and heteroscedastic models.
- 2) Refer to the graphical representations and patterns in residuals that indicate autocorrelation given in Section 5.6
- 3) Reflect on how a zero-intercept model alters the interpretation of coefficients and affects the model's fit and predictions compared to models with intercepts.
- 4) Refer to Sub-Section 5.8.1 and answer.



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UNIT 6 ESTIMATION OF MULTIPLE REGRESSION MODELS*

Structure

- 6.0 Objectives
- 6.1 Introduction
- 6.2 Specification of the Model
 - 6.2.1 Matrix Representation
 - 6.2.2 Classical Assumptions
- 6.3 OLS Method of Estimation
 - 6.2.1 Mean of OLS Estimators
 - 6.2.2 Variance of OLS Estimators
 - 6.2.3 Estimator of Error Variance
- 6.4 Properties of OLS Estimators
 - 6.4.1 Algebraic Properties
 - 6.4.2 Statistical Properties
- 6.5 Best Linear Unbiased Estimator (BLUE)
- 6.6 Let Us Sum Up
- 6.7 Key Words
- 6.8 Some Useful Books
- 6.9 Answers/Hints to Check Your Progress Exercises

6.0 OBJECTIVES

After going through this Unit you should be in a position to

- explain the concept of multiple regression models with two or more explanatory variables;
- represent multiple regression model in matrix form;
- apply ordinary least squares (OLS) method to find out the estimators;
- describe the properties of the OLS estimators; and
- prove that OLS estimators are the best linear unbiased estimator (BLUE).

6.1 INTRODUCTION

In Unit 4 we discussed the relationship between two variables (one explained, one explanatory) using the simple regression model. We considered the dependent

variable Y to be a function of the independent variable X . However, in practise we notice that there is more than one explanatory variable in a regression model. In such situations it is difficult to draw conclusion on how X affects Y *ceteris paribus* as it is unrealistic to assume that all other factors influencing Y are not correlated to X .

Multiple regression models provide an amicable solution to such a problem. As the name suggests, we consider two or more explanatory variables (and one dependent variable) in multiple regression models. Examining the effects of a particular independent variable on the dependent variable *ceteris paribus* becomes possible as with multiple regression models we can control for several other factors simultaneously which could have influence on the dependent variable. This enables us to test economic theories as well as evaluate policy impacts. Several explanatory variables can be accommodated in multiple regression models allowing us to draw inferences on causality. Multiple regression models enable us to add suitable factors to the model resulting in explanation of more variation in the dependent variable. Such benefits of a multiple regression model make it one of the most widely used empirical tools for analysis in economics and social sciences. In the following section we will develop an understanding about a multiple regression model and evaluate the advantages of using it for analysis.

6.2 SPECIFICATION OF THE MODEL

In multiple regression models we have one explained variable and many explanatory variables (say, k). The multiple regression model is specified as follows:

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i \quad (i = 1, 2, \dots, n) \quad \dots (6.1)$$

Here, the subscript ' i ' denotes the observation number in the sample, while the subscript ' k ' denotes the explanatory variable. Thus, Y_i is the i^{th} observation of the dependent variable, $\beta_1, \beta_2, \dots, \beta_k$ are the parameters, $X_{1i}, X_{2i} \dots X_{ki}$ are the i^{th} observations on the explanatory variables, and u_i is the stochastic error term. Specification of the multiple regression model is similar to that of the two-variables model, except that there is more than one explanatory variable here. Generally, we assume the first explanatory variable to be unity ($X_1 = 1$) for all the observations. Under such an assumption, the multiple regression model becomes

$$Y_i = \beta_1 + \beta_2 X_{2i} + \beta_3 X_{3i} + \dots + \beta_k X_{ki} + u_i \quad (i = 1, 2, \dots, n) \quad \dots (6.2)$$

The first parameter, β_1 , in (6.2) is the intercept that we discussed in Unit 4. To avoid confusion in notation, very often we drop the superscript ' i ', and represent multiple regression model in the following form:

$$Y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + u \quad \dots (6.3)$$

As in the case of two-variable regression model, the multiple regression model is estimated by the ordinary least squares (OLS), method of moments, and the maximum likelihood method. We will discuss the OLS and maximum likelihood methods of estimation in this Unit. In both the methods, the approach is the same as mentioned in Unit 4. In the OLS method, for example, we must minimise the

sum of squared residuals ($\sum \hat{u}_i^2$) and take partial derivatives with respect to the parameters. Thus, there are k normal equations (see Unit 4) for a regression model with k independent variables. Finding equilibrium estimates of all the parameters in the model is a difficult and time-consuming process. In order to simplify the process of finding equilibrium values, we take the help of matrix algebra.

6.2.1 Matrix Representation

In Unit 3 of this course, we provided an overview of matrix algebra. Let us express equation (6.2) as follows:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U} \quad \dots (6.4)$$

where

$$\begin{array}{cccc} \mathbf{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} & \mathbf{X} = \begin{bmatrix} X_{11} & X_{12} & \dots & X_{1k} \\ \vdots & \vdots & \dots & \vdots \\ X_{n1} & X_{n2} & \dots & X_{nk} \end{bmatrix} & \boldsymbol{\beta} = \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} & \mathbf{U} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \\ (n \times 1) & (n \times k) & (k \times 1) & (n \times 1) \end{array}$$

The dependent variable $\mathbf{Y}$ is a $(n \times 1)$ vector. The explanatory variables are denoted by matrix $\mathbf{X}$ which has the order $(n \times k)$. As you can see, i^{th} row indicates the i^{th} observation and j^{th} column indicates j^{th} explanatory variable. Thus, X_{ij} represents the value of the j^{th} explanatory variable of the i^{th} observation. Vector $\boldsymbol{\beta}$ denotes the set of explanatory variables. As there are k explanatory variables in the model, the order of $\boldsymbol{\beta}$ vector is $(k \times 1)$. The vector $\mathbf{U}$ denotes the stochastic error variable, which has an order of $(n \times 1)$. These notations are commonly used in econometrics. You should keep track of the dimensions of the matrices.

6.2.1 Classical Assumptions

Recall the four assumptions regarding the error variable (u). These are: (i) $E(u_i) = 0$, (ii) $E(u_i^2) = \sigma^2$, (iii) $E(u_i u_j) = 0$ for $i \neq j$, and (iv) $E(X_i u_i) = 0$. We need to have corresponding assumptions for the multiple regression model. The errors are identically and independently distributed with mean zero and variance σ^2 . The explanatory variables are non-stochastic and independent of the error terms. The explanatory variables are linearly independent. Thus,

These assumptions are

$$(i) \quad E(\mathbf{U}) = \mathbf{0} \quad \dots (6.5)$$

Assumption (6.5) says that $E(u_i) = 0$; $i = 1, 2, \dots, n$. The expected value of u_i is zero.

$$(ii) \quad E(\mathbf{U}\mathbf{U}') = \sigma^2 \mathbf{I} \quad \dots (6.6)$$

Assumption (6.6) is a precise way of expressing two assumptions: $E(u_i u_j) = 0$ and $E(u_i^2) = \sigma^2$. Let us define the variance-covariance matrix $\mathbf{U}\mathbf{U}'$. Note that it has a dimension of $(n \times n)$. It is given by

$$\begin{bmatrix} \text{var}(u_1) & \text{cov}(u_1 u_2) & \dots & \text{cov}(u_1 u_n) \\ \vdots & \vdots & \dots & \vdots \\ \text{cov}(u_n u_1) & \text{cov}(u_n u_2) & \dots & \text{var}(u_n) \end{bmatrix} = \begin{bmatrix} \sigma^2 & 0 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \sigma^2 \end{bmatrix}$$

In the above matrix, the diagonal elements are σ^2 and the off-diagonal elements are zero.

$$(iii) \quad X \text{ has full rank} \quad \dots (6.7)$$

It implies that there is no exact relationship among the explanatory variables. Thus, none of the explanatory variables is a linear combination of other explanatory variables. If there is an exact relationship between any two explanatory variables, there will be a decline in the rank of matrix X . The rank of matrix X is and $k < n$. An implication of assumption (6.7) is that the matrix $X'X$ is invertible, that is, $(X'X)^{-1}$ exists.

$$(iv) \quad E(X'U) = 0 \quad \dots (6.8)$$

Assumption (6.8) implies that none of the explanatory variables is related with the error variable u . In other words, the error variable U cannot be predicted on the basis of the independent variables. In order to satisfy this assumption, we assume that X is non-stochastic (i.e., fixed in repeated samples). In Unit 13 we will relax this assumption and assume that some of the regressors are stochastic in nature.

By assumption $\text{var}(u_1) = \text{var}(u_2) = \dots = \text{var}(u_n) = \sigma^2$. Thus, variance of the error term is constant across all the observations. This feature of the error term is called homoscedasticity. When the error variance does not remain constant across observations (that is, $\text{var}(u_1) \neq \text{var}(u_2) \neq \dots \neq \text{var}(u_n)$) we have a situation of heteroscedasticity, which we will discuss in Unit 11. Similarly, we assumed that $E(u_i u_j) = 0$. It implies that the error term of one observation is not correlated with the error term of another observation. In other words, there is no autocorrelation.

6.3 OLS METHOD OF ESTIMATION

In the two-variable case we minimised the residual sum of squares (RSS), given by $\sum \hat{u}_i^2$. In the multiple regression model, we have to minimise the RSS for obtaining the estimators. We mentioned above in equation (6.4) that the regression model is given by

$$Y = X\beta + U$$

The estimated regression model is given by

$$\hat{Y} = X\hat{\beta} \quad \dots (6.9)$$

Therefore,

$$\hat{U} = Y - X\hat{\beta} \quad \dots (6.10)$$

The RSS is given by $\hat{U}'\hat{U}$ which has a dimension of $(1 \times n)(n \times 1) = (1 \times 1)$. Thus, RSS is a scalar quantity.

From equation (6.10) we have

$$\hat{U}'\hat{U} = (Y - X\hat{\beta})'(Y - X\hat{\beta}) \quad \dots (6.11)$$

On simplification, (6.11) can be written as

$$\hat{U}'\hat{U} = Y'Y - \hat{\beta}'X'Y - Y'X\hat{\beta} + \hat{\beta}'X'X\hat{\beta}$$

Since $\hat{\beta}'X'Y$ and $Y'X\hat{\beta}$ are scalar quantities (the order of both the matrices is (1×1)), we can write

$$\hat{U}'\hat{U} = Y'Y - 2\hat{\beta}'X'Y + \hat{\beta}'X'X\hat{\beta} \quad \dots (6.12)$$

In the least squares method we choose such value of $\hat{\beta}$ that minimises the **RSS**. The conditions of finding a minimum are: (i) first derivative should be equal to zero, and (ii) second derivative should be positive.

Let us take the derivative of $\hat{U}'\hat{U}$ with respect to $\hat{\beta}$.

$$\frac{\partial \hat{U}'\hat{U}}{\partial \hat{\beta}} = 0 - 2X'Y + 2X'X\hat{\beta} \quad \dots (6.13)$$

While taking derivative of (6.12) with respect to $\hat{\beta}$, you should pay special attention to derivative of the term $\hat{\beta}'X'X\hat{\beta}$. Both $\hat{\beta}'$ and $\hat{\beta}$ are present here. We have taken its derivative is $2X'X\hat{\beta}$, and not $2\hat{\beta}'X'X$. Let us look into the following example.

Example 6.1

Suppose we have two vectors: **a** and **x**.

$$a = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Thus,

$$a'x = a_1x_1 + a_2x_2 + a_3x_3$$

Let us take derivative of $a'x$ with respect to x .

$$\frac{\partial a'x}{\partial x} = \frac{\partial a'x}{\partial \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \frac{\partial}{\partial \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} a_1x_1 + a_2x_2 + a_3x_3 = \begin{bmatrix} a_1 + 0 + 0 \\ 0 + a_2 + 0 \\ 0 + 0 + a_3 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = a$$

Coming back to equation (6.13), we eliminate $\hat{\beta}'$ (and not $\hat{\beta}$) while taking the derivative of $\hat{U}'\hat{U}$ with respect to $\hat{\beta}$. By equating $\frac{\partial \hat{U}'\hat{U}}{\partial \hat{\beta}} = 0$, we obtain

$$-2X'Y + 2X'X\hat{\beta} = 0$$

Or,

$$X'X\hat{\beta} = X'Y \quad \dots (6.14)$$

Equation (6.14) represents the 'normal equations' of the least squares method.

If we pre-multiply equation (6.14) by $(X'X)^{-1}$, we obtain

$$\hat{\beta} = (X'X)^{-1}X'Y \quad \dots (6.15)$$

Let us take the second derivative of $\hat{U}'\hat{U}$ with respect to $\hat{\beta}$. It is similar to taking the derivative of (6.13). Thus,